1. a) Starting Address = Base Address

If G = 0, Ending Address = Base Address + Limit

If G = 1, Ending Address = Base Address + Limit * 4K

Index

045 H

S.A. = B.A. = 23308810 H

G = 1, E.A. = 23308810 + 18BD7FFF H

= 3BEE080F H

∴ 92A15BF8 H does not belong in this range.

Index

1A3 H

S.A. = B.A. = 4FE4DBB8 H

G = 0, E.A. = 4FE4DBB8 + 2345A H

= 4FE71012 H

∴ 92A15BF8H does not belong in this range.

Index

2F6 H

S.A. = B.A. = 866125A3 H

G = 1, E.A. = 866125A3 + 1132FFFF H

= 979425A2 H

∴ 92A15BF8H belongs in this range.

CamScanner

So, index number = 2F6 H

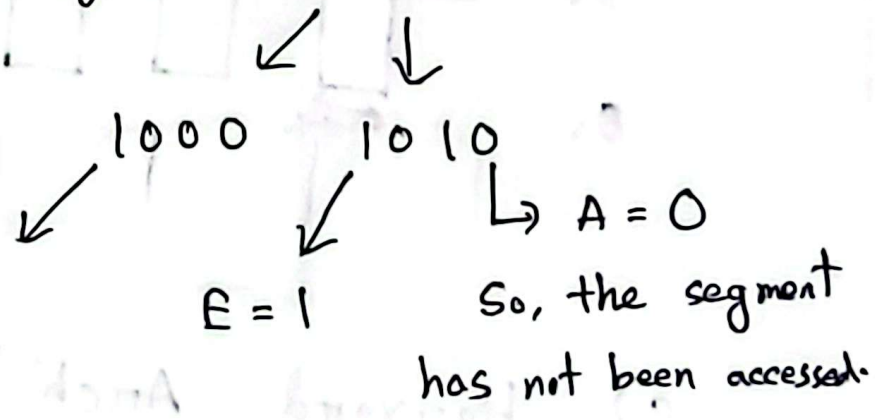
b) P.A. = B.A. + offset

∴ offset = P.A. − B.A.

= 867003A0 − 8661253A3 H

= EDDFD H

c) Access Rights = 8A H



d) E = 1, So, Code Segment.

e) P = 1, So, the descripton is defined.

2. a) DAC = 10 bit $\simeq 2^{10}$ levels

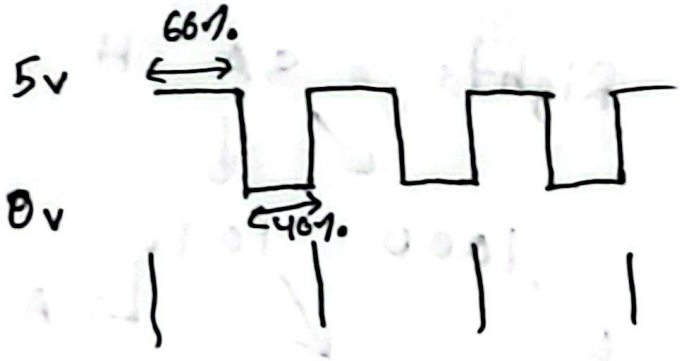
(0 – 1023)

100 % duty Cycle → AnalogWrite(1023)

∴ 1 % " → " $\left(\frac{1023}{100\%}\right)$

∴ 60 % " → " $\left(\frac{1023 \times 60\%}{100\%}\right)$

$= 613.8 \simeq 614$



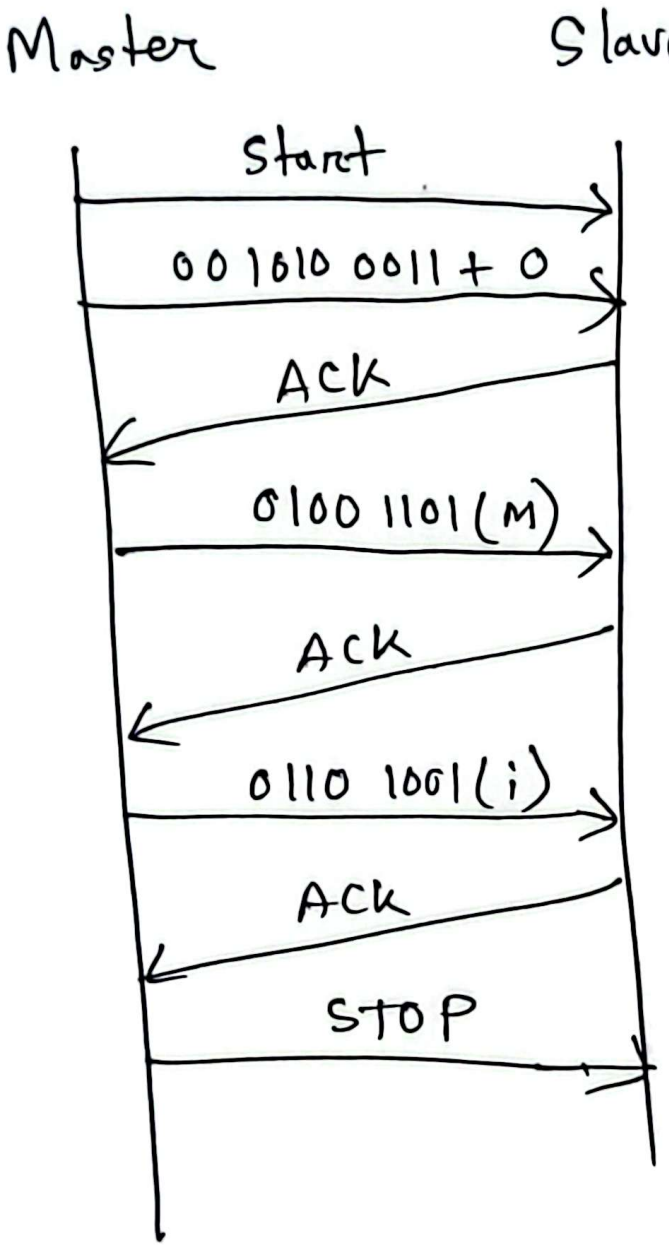
c) Harvard Architecture

because → 1. Speed priority.

2. No cost issues.

3. No complexity "

b)



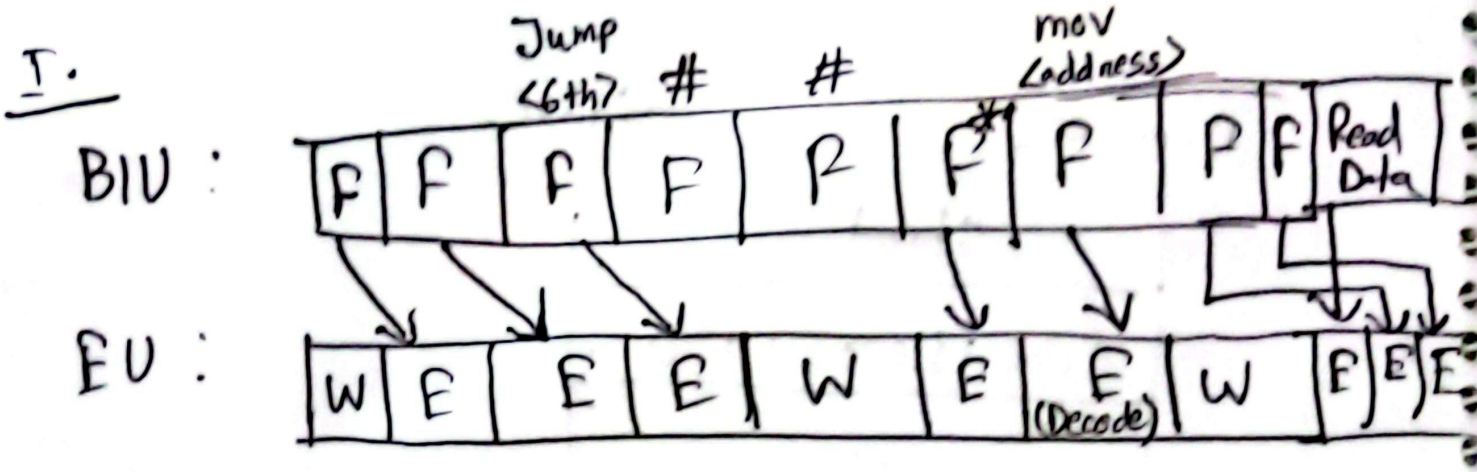
M = 77

i = 105

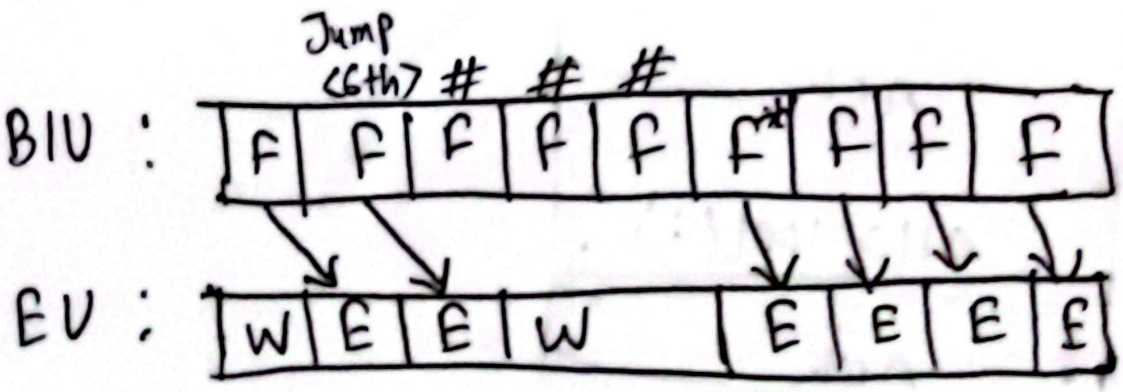
CamScanner

3. a)

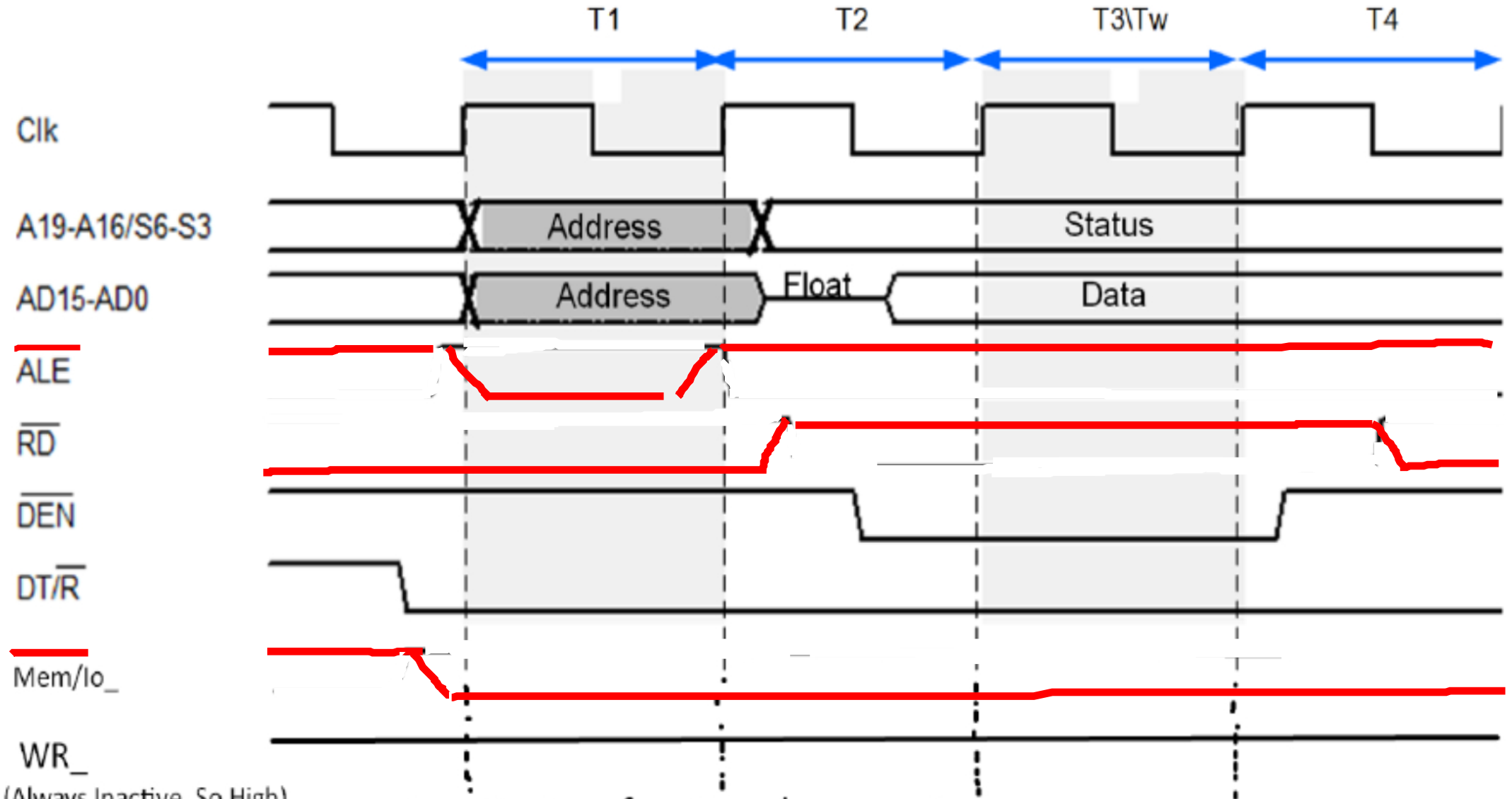
I.



II.



discard



Bus timing for Read operation

Activate Windows

Q4

a)

LDA FH
ADD DH
ADD FH
ADD DH
SUB BH
OUT
HLT

b)

Address	Content		Address	Content
0H	0011 1111		8H	XXXX XXXX
1H	0001 1101		9H	XXXX XXXX
2H	0001 1111		AH	XXXX XXXX
3H	0001 1101		BH	0000 0101 (5)
4H	0010 1011		CH	0001 0010 (18)
5H	1101 XXXX		DH	0000 0001 (1)
6H	1011 XXXX		EH	0000 0110 (6)
7H	XXXX XXXX		FH	0000 0011 (3)

c) T4: 1010 0011 0011 (A33)
T5: 1110 0000 0010 (E02)
T6: 1100 0001 1111 (C1F)